

Class Test
Subject: Fluid Mechanics
3rd Semester CE (Sec.B)

Date: 08.11.2025

Total Marks: 30

Q.1 Derive Euler's equation of motion along a stream for an ideal fluid stating clearly the assumptions. Explain how this is integrated to get Bernoulli's equation along a stream line. [8+2=10]

Q.2 A fluid flow field is given by $\vec{V} = x^2 y \hat{i} + y^2 z \hat{j} - (2xyz + yz^2) \hat{k}$. Prove that it is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point (2, 1, 3). [2+8=10]

Q.3 (a) What are the convective and local acceleration? Define velocity potential function and stream function. [05]

(b) Water is flowing through a pipe of 100 mm diameter under a pressure of 19.62 N/cm² (gauge) and with mean velocity of 3 m/s. Find the total head of the water at a cross-section, which is 8 m above the datum line. [05]